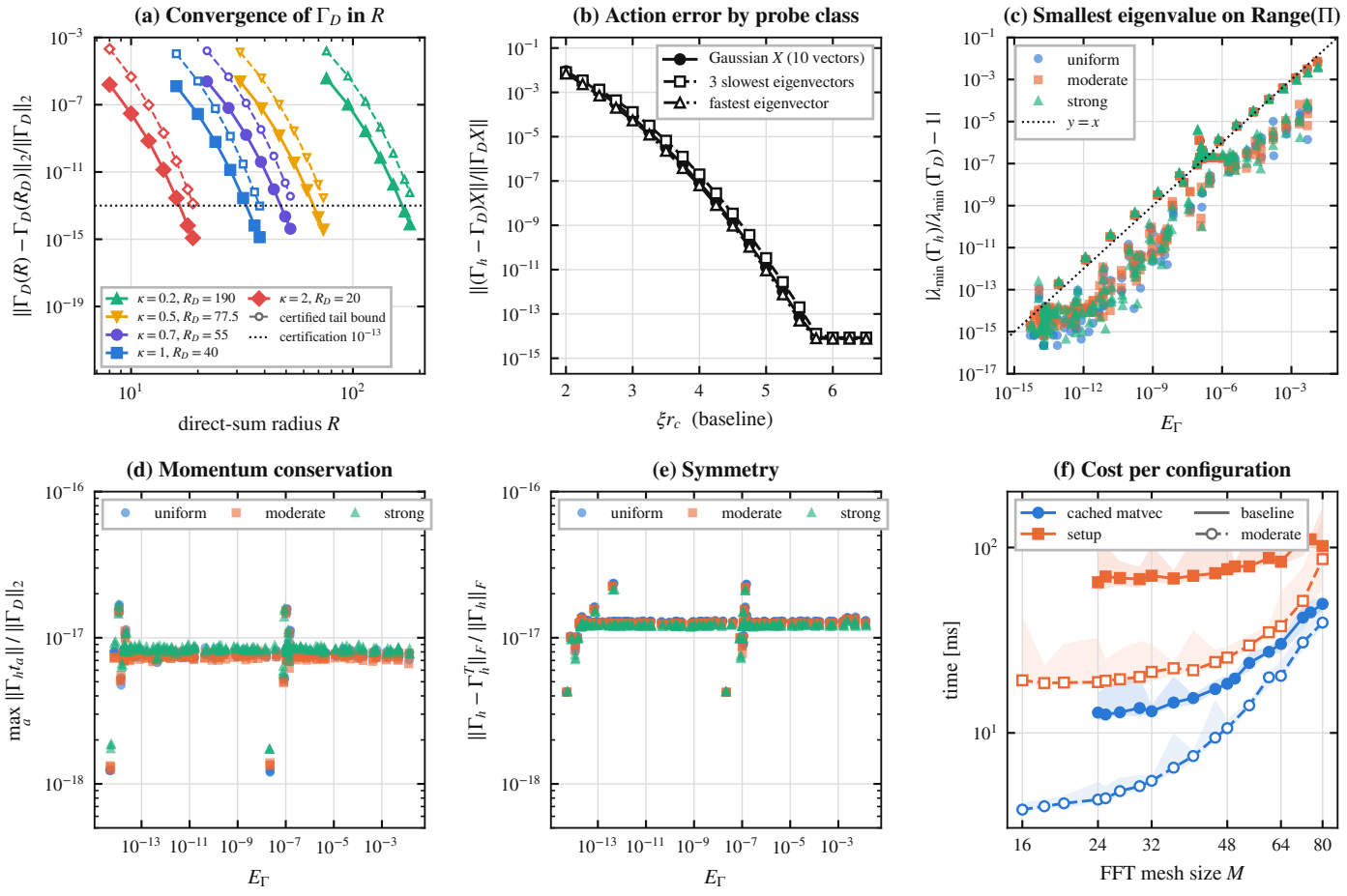

(a) Convergence of $\Gamma_D$ in $R$
$\|\Gamma_D(R) - \Gamma_D(R_D)\|_2 / \|\Gamma_D\|_2$
direct-sum radius $R$
$\kappa = 0.2$, $R_D = 190$
$\kappa = 0.5$, $R_D = 77.5$
$\kappa = 0.7$, $R_D = 55$
$\kappa = 1$, $R_D = 40$
$\kappa = 2$, $R_D = 20$
certified tail bound
certification $10^{-13}$
(b) Action error by probe class
$\|(\Gamma_h - \Gamma_D)X\| / \|\Gamma_D X\|$
$\xi r_c$ (baseline)
Gaussian $X$ (10 vectors)
3 slowest eigenvectors
fastest eigenvector
(c) Smallest eigenvalue on Range($\Pi$)
$|\lambda_{\min}(\Gamma_h)/\lambda_{\min}(\Gamma_D) - 1|$
$E_\Gamma$
uniform
moderate
strong
$y = x$
(d) Momentum conservation
$\max_a \|\Gamma_h t_a\| / \|\Gamma_D\|_2$
$E_\Gamma$
uniform
moderate
strong
(e) Symmetry
$\|\Gamma_h - \Gamma_h^T\|_F / \|\Gamma_h\|_F$
$E_\Gamma$
uniform
moderate
strong
(f) Cost per configuration
time [ms]
FFT mesh size $M$
cached matvec
setup
baseline
moderate